T 13

$n_1 = 139$ $\quad$ $n_2 = 1000$

$S_{1L} = 5.722 = a_1$ $\quad$ $S_{2L} = 6.161 = b_1$

$S_{1W} = 4.612 = a_2$ $\quad$ $S_{2W} = 5,055 = b_2$

$H_0: \sigma_1^2 = \sigma_2^2$

$H_1: \sigma_1^2 \neq \sigma_2^2$

$\frac{(n_1-1) S_x^2}{a_1^2} \sim \chi^2(n_1-1)$ $\quad$ $\frac{(n_2-1) S_y^2}{b_1^2} \sim \chi^2(n_2-1)$

$$\Delta = \frac{S_x^2 / a_1^2}{S_y^2 / b_1^2} \sim F(n_1-1, n_2-1)$$

$P(\Delta \le C_1 | H_0) = \frac{\alpha}{2}$ $\quad$ $C_1 = \cancel{0,765}\ 0,767$

$P(\Delta > C_2 | H_0) = \frac{\alpha}{2}$ $\quad$ $C_2 = 1,303$

$G: \tilde{\Delta} \le F_{\frac{\alpha}{2}}(n_1-1, n_2-2)$ $\quad$ $\tilde{\Delta} \ge F_{1-\frac{\alpha}{2}}$ $\quad$ $\tilde{\Delta} = \frac{S_x^2}{S_y^2} = \left(\frac{4.612}{5,055}\right)^2 = 0.832$

$P\text{-value} = 2P(\Delta \ge \tilde{\Delta})$

$0,767 < 0,832 < 1,303$ ✓

Нет оснований отвергнуть H_0 для ширины

$\tilde{\Delta} = \left(\frac{5,722}{6,161}\right)^2 = 0,862$

$0,767 < 0,862 < 1,303$ ✓

Нет осн. отвергнуть H_0 для длины

$\cancel{W = P(|\Delta|}$

$$W = P(C_1 \le \Delta \le C_2 | H_0) =$$

$$= P\left(\frac{S_x^2}{S_y^2} < \frac{\sigma_1^2}{\sigma_1^2} C_1 \Big| H_0\right) + P\left(\frac{S_x^2}{S_y^2} > C_2 \frac{\sigma_1^2}{\sigma_1^2} \Big| H_0\right)$$

ТИ $\quad H_0: a = b \quad H_1: a > b$

$$\bar{x} = \frac{-1,11 - 6,10 + 2,42}{3} = -1,596$$

$$\bar{y} = -2,6$$

$$\frac{(\bar{x} - \bar{y}) - (a - b)}{\sqrt{\frac{\sigma_x^2}{n} + \frac{\sigma_y^2}{m}}} \overset{\text{при } H_0}{=\!=} 0,929$$

$$\sim N(0, 1)$$

$$u_{1-\alpha} = 1,6449$$

$0,929 < 1,645$ — нет осн. отвергнуть H_0

$$p\text{-value} = P\left(\frac{\bar{x} - \bar{y}}{\sqrt{\frac{\sigma_x^2}{n} + \frac{\sigma_y^2}{m}}} > 0,929 \Big| H_0\right) =$$

$$= 1 - \underbrace{\Phi(0,93)}_{\substack{\text{норм. распр} \\ = 0,82}} = 0,18 > \alpha$$

нет осн. отвергнуть H_0

$$W = P\left(\frac{\bar{x} - \bar{y} - (a - b)}{\sqrt{\frac{\sigma_x^2}{n} + \frac{\sigma_y^2}{m}}} > 1,645 \Big| H_1\right) =$$

$$= 0,237$$

$\cancel{W = P(}$

$$W(\Delta) = 1 - \Phi\left(1{,}645 - \frac{\overset{a-b}{=}\Delta}{\sqrt{\frac{\sigma_x^2}{n} + \frac{\sigma_y^2}{m}}}\right)$$

$$W(\Delta) = 1 - \Phi(1{,}645 - 0{,}926 \cdot \Delta)$$